FERHAT TÜFEKÇİ

π-Shaped Computer Engineer | Expert Business Analyst (Systems Engineer) | Agile Full Stack Software Development & Testing Specialist

Contact: +90 507 600 81 79 | Turkey | ferhattufekci@outlook.com.tr | set up a call | My Website | All My Links | Türkçe Özgeçmişim (My Resume in Turkish Language)

Languages: Turkish (Native) | English (Graduated from Ankara University, which provides education in 100% English)

Personal Security Clearance: Valid until 12.04.2026

Driving Licence: B-Class license; 100,000+ km of driving experience

Military Service: Completed

SUMMARY

The opportunity to solve a common problem that many people experience — using just a computer in my room — shaped both my education and career. I hold a strong academic background in computer science, having completed my high school, associate, and bachelor's degrees with high honors. Since 2011, I've built versatile experience across the Software - Systems Development Lifecycle (SDLC) — taking on roles such as Software Engineer, Systems Engineer (Business Analyst), and Test Engineer. I have contributed to a wide range of projects, from small-scale applications to large, highly available, and auto-scalable enterprise solutions. After more than five years of experience in the defense industry (at Havelsan and MilSOFT) and private sector, I transitioned back to freelance work in May 2023. Today, I wear multiple hats as a Full Stack Agile Developer, IT Business Analyst (Systems Engineer), Agile Tester, and Blogger. Throughout my career, I have continuously developed not only my technical skills but also my capabilities in project management, team leadership, and software architecture. I have taken on responsibilities in leading Agile teams, making critical technical decisions, and designing long-term, sustainable software solutions—strengthening my strategic competencies in the process. As a professional who can manage software development processes at both operational and architectural levels, I prioritize aligning team coordination with a strong technical vision. I'm a collaborative and enthusiastic team player who values integrity, continuous improvement, and delivering successful outcomes.

SKILLS

Business Analyst / Systems Engineer Skills

- Proficient in requirements analysis and management processes in accordance with INCOSE guidelines and industry standards.
- Competent in preparing Technical Spesifications, Mockups, SSS, SRS, STP, STD, STR, and User Manual documents in accordance with MIL-STD, STANAG standards, and HAVELSAN HVL-TEMP-SRS, MilSOFT document templates.
- Experienced in Agile project management using the Scrum Framework.
- Adept at using tools like Jira, MS Project, Confluence, Doors, Microsoft Office Apps., Enterprise Architect, and Figma.

Testing Skills

- Experienced in implementing ISTQB-recommended practices for both automation and manual testing within Agile environments.
- Proficient in automating end-to-end testing processes using tools such as Selenium, Postman, Swagger, and Cucumber, and applying test automation techniques like the Page Object Model (POM).

Full-Stack Software Development Skills

- Proficient in writing clean, maintainable, and efficient code.
- Experienced in layered & monolith & microservices architectures, procedural programming, object-oriented programming (OOP), multithreading programming, test driven development (TDD), SOLID principles, design patterns, data structures, algorithms (searching/sorting), and debugging.
- Skilled in version control and CI/CD processes using tools like SVN, Git, GitHub, GitHub Actions, Bitbucket, Maven, Gradle, Jenkins, and SonarQube.
- Experienced in container-based application development with Docker.
- Capable of creating user-friendly web interfaces using HTML, CSS, JavaScript, React.js, and Bootstrap.
- Experienced in relational database (RDMS) design and management with SQL, Transact-SQL (T-SQL), Microsoft SQL Server, and PostgreSQL.
- Proficient in application development using UML, C++, C# (Windows Form, ASP.Net), Python, and Java (Swing, JavaFX, Spring Framework, Spring Boot, JUnit, Hibernate).

EDUCATION

Lessons Learned

- Miuul, **Data Science & Machine Learning Bootcamp Participant**, GitHub, Certificate — **05.2023 - 11.2023**

  It is an online-live and mentor-supported Data Scientist and Artificial Intelligence (AI) Engineer Training program. In this program, I developed projects that solved more than 10 real-life problems.
- Ankara University, **B.E.N.G. Computer Engineering (in %100 English),** GPA: 3,01 / 4.00 — **2014 - 2019**
- Gazi University, **F.D. Computer Programming,** GPA: 3.25 / 4.00, *Top Score of Department* — **2011 - 2013**
- Ahmet Çiçek Technical High School, **Web Technologies and Software,** GPA: 79 / 100, *Top Score of Department* — **2007 - 2011**

EXPERIENCE

10+ Years of Professional Experience

FREELANCE

06.2010 - 04.2021 & 05.2023 - Present

Business Analyst (Systems Engineer) & Full Stack Agile Developer & Agile Tester & Blogger, Medium

- As a systems engineer proficient in INCOSE-aligned practices, I analyze and manage stakeholder needs to ensure that software is not only built right, but also built for the right reasons. I drive clarity across all stages of the project by eliciting, documenting, and validating requirements — maintaining traceability and alignment with business goals.
- As an ISTQB-oriented Agile Tester, I design and implement structured, risk-based test strategies that ensure comprehensive verification and validation across all stages of the software lifecycle — confirming that the product is not only built correctly but also fulfills real user needs. I validate both functional and non-functional requirements, assess usability, and ensure robust performance under real-world conditions. I conduct root cause analysis to identify underlying issues, create clear and traceable defect reports, and manage the defect lifecycle efficiently to support timely resolution, reduce rework, and drive continuous quality improvement.
- As a full stack agile developer, I translate requirements into scalable, maintainable code — delivering robust applications using modern technologies. I also design and maintain automated regression suites, reducing manual effort, accelerating feedback loops, and increasing release confidence.
- I regularly share technical insights and personal reflections through my Medium blog, contributing to the community.

Used Technologies: Scrum, ISTQB Foundation Level, Microservice, Monolith, Jira, Trello, Confluence, UML, moqups.com, excalidraw.com, Figma, Maven, Jenkins, SonarQube, Git, Gradle, Bitbucket, Docker, Bash, Java, Java Spring Framework, JUnit, Html, CSS, JavaScript, Bootstrap, React.js, PHP, Laravel, Hibernate, SQL, PostgreSQL, Postman, Cucumber, Selenium, Intellij IDEA, Microsoft Office Programs.

MilSOFT

02.2023 - 05.2023

Senior Systems & Test Engineer

MilSOFT is the only company in the **Turkish Defense Industry** to reach the **CMMI-5** level with 100% coverage in all of its projects.

Mil-DLP (MilSOFT Multi Data Link Processor) is a system that provides multi-link processing capability to share tactical information among surface, subsurface, air, and land platforms/units.

I took tasks on the Mil-DLP Product Line (DLP of Aselsan company's Siper & Hakim projects are products of this product line) which was developed by applying Agile Methodology Scrum Framework.

- Created Organizational Memory with the information acquired during the orientation, and the contribution time of those who joined the project later was shortened.
- Planned and monitored test activities in the Sprint.
- Designed and created test cases for **NATO Tactical Data Link Systems** such as Link 11, Link 16, and Link 22 with 100% coverage among **6,000 pages of STANAG** requirements.
- Performed manual and automation tests.
- Verified the intermediate product at the end of the sprint.
- Performed impact analysis for errors, deficiencies, and improvements found during manual and automation tests.
- Created the Problem Report record on Jira by checking whether a bug report was opened before and if a bug report was opened before, whether it was opened correctly (severity, priority, impact analysis, component, etc. fields).
- Performed verification and regression tests of the Problem Report/Features whose status is Testing Request.

Used Technologies: Scrum, CMMI-5, ISTQB Foundation Level, STANAG, Jira, Confluence, Doors, STANAG, UML, SVN, Git, Bitbucket, Jenkins, SonarQube, Specific software developed by Milsoft for Automation Test Case Writing, Gateway, Microsoft Office Programs.

HAVELSAN

04.2021 - 02.2023

Senior Full-Stack Agile Developer & Business Analyst (Systems Engineer) & Agile Tester

HAVELSAN, ranked #99 in DefenseNews 2020, is a leading defense software company specializing in large-scale, mission-critical systems.

I contributed **end-to-end** in the MBSM Web Application, a strategic information system developed with **Microservice** Architecture and Agile Scrum, built on HAVELSAN's **national ERP platform**, KOVAN.

- **During the analysis phase**, according to the **MIL-STD-408** software development standard and HVL-TEMP-SRS Havelsan document standard, I took tasks of creating, managing, reporting on, verifying, and validating the requirement set consisting of approximately **2.500** software requirements for **5 Departments** of the Ministry of National Defense.
- **During the design and development phases,** I took the tasks of designing and developing the application to meet the defined requirements, writing unit tests, fixing bug records, and reviewing Pull Requests.
- **During the testing phase,** I took part in the early detection of errors with the shift-left approach and reducing the cost of error correction/faulty development, planning the tests to be performed in the sprint, designing the test scenario with 100% coverage among 2.500 requirement sets, creating the test procedure document, publishing the approved test procedures on Jira, running static/dynamic, functional/nonfunctional tests, creating and tracking the records of the errors/deficiencies/improvements found by performing impact analysis on Jira, presenting the status of the tests in status meetings, standardizing the error records on Confluence to avoid creating duplicate/faulty records, and automating manual tests to reduce the cost of regression testing.

Used Technologies: Scrum, CMMI-3, ISTQB, MIL-STD-408, Microservice, Jira, Confluence, Doors, UML, Git, Gradle, Bitbucket, Maven, Jenkins, SonarQube, Docker, Bash, Java, Java Spring, JUnit, JSF, PrimeFaces, Hibernate, SQL, PostgreSQL, Postman, Cucumber, Selenium.

MINISTRY OF NATIONAL EDUCATION **09.2019 - 01.2021**

Information Communication Technology Teacher & TUBITAK 4006 Project Manager

- I ensured that 26 different projects that got secondary school students interested in software in the field of information technologies were completed on time simultaneously.
- I ensured the establishment of the school library.

Used Technologies: Scrum, Trello, WordPress, Html, CSS, JavaScript, Ardunio, Sensors, Microsoft Office Programs.

CERTIFICATES

- **07.2025, Master Microservices with Spring Boot, Cloud, and Docker** Certificate
  Udemy
- **07.2025, Product Owner** Certificate
  BTK Academy
- **04.2025, Advanced Systems Engineering** Certificate
  Ömer Ertekin - PSConsultech
- **01.2025, Scrum** Certificate
  BTK Academy
- **08.2023, Machine Learning** Certificate
  Miuul
- **07.2023, Feature Engineering** Certificate
  Miuul
- **07.2023, Python Programming for Data Science** Certificate
  Miuul
- **05.2023, Software Test Automation** Certificate
  BTK Academy
- **05.2023, API Testing** Certificate
  BTK Academy
- **02.2023, Amazon Web Services (AWS)** Certificate
  Udemy
- **09.2022, Jenkins** Certificate
  BTK Academy
- **05.2022, Docker** Certificate
  Udemy
- **05.2015, Catia** Certificate
  Gazi University

HOBBIES & INTERESTS & VOLUNTARY ACTIVITIES

- INCOSE Turkey – Volunteer Writer (Creating volunteer content on topics such as systems engineering, digital twins, and lifecycyle process. Contributing to the development of technical and community-oriented materials to raise awareness of INCOSE Turkey's mission.)

  Selected contributions include:
  - Digital Twin in the Defense Industry: The Key to Agility in the System Life Cycle
  - Writing Requirements from Scratch: A Systematic Process with the INCOSE Approach
  - …and more.
- I am a member of the Ankara Club Association, which was established with the directives of Mustafa Kemal ATATÜRK.
- I enjoy contributing as a support player in team-based games such as League of Legends, where communication and strategy are essential.
- I don't smoke. At the same time, I am a volunteer member of Yeşilay.
- I am a Volunteer Member of the Disaster and Emergency Management Presidency (AFAD).
- I enjoy trekking and nature walks as a way to stay active, focused, and connected with the environment.

REFERENCES

- **Can TUNÇ - Lobi Bilişim Project Manager** +90 530 527 85 89
  (*worked together for 1 year in MBSM Project)*
- **Doğan VARLI - Havelsan Project Manager** dvarli@havelsan.com.tr, +90 539 552 42 39
  (*worked together for 1 year in MBSM Project)*
- **İbrahim Cihan YETİŞKEN - Head of Department Gazi Univesity TUSAŞ – KAZAN VOLATIONAL HIGH SCHOOL**
  (*worked together for 2 years)* cyetisken@gazi.edu.tr, +90 553 629 83 42
- **Berkcan ÇELİK- Software Engineer at Yapı Kredi Technology** berkcan.celik@yteknoloji.com.tr, +90 554 559 74 04
  (*worked together for 1 year in MBSM Project)*